Enterprise Network Design for KtmTechnologies

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1 Introduction

This proposal presents a compact enterprise network for **KtmTechnologies**, a technology and electronics organization with a corporate headquarters, an R&D department, a production and logistics plant, a data center and a branch sales office. The design uses 10 internal routers, departmental LANs and VLANs, 14 point-to-point links, and a redundant core.

The private address block `10.0.0.0/16` is subnetted using VLSM. Internal routing uses a three-area OSPF hierarchy: Area 0 is the backbone, Area 1 serves the headquarters and commercial departments, and Area 2 serves R&D, production, logistics and the branch. A default route leads to the ISP, which uses a static return route to the enterprise network.

2 Network Topology

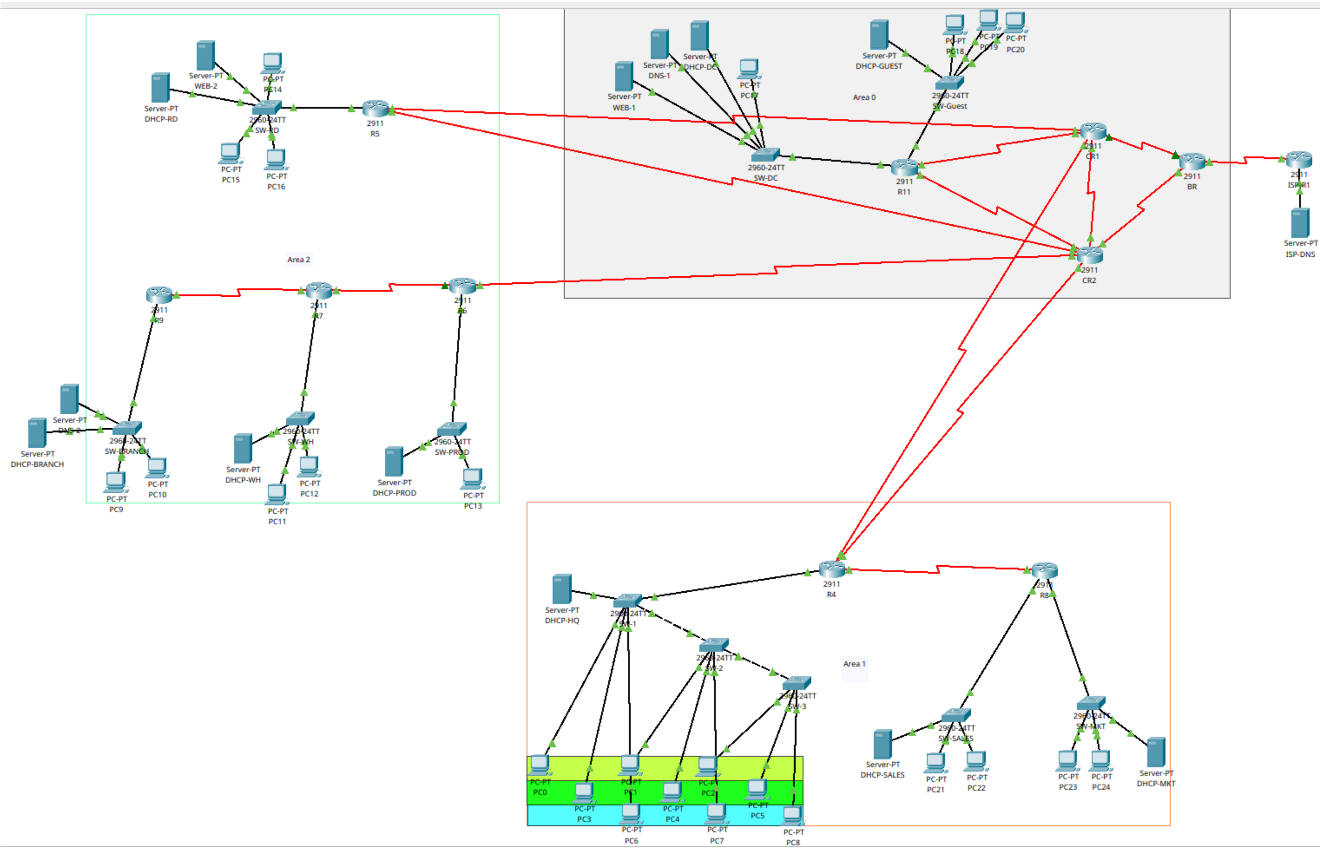

Figure 1: Proposed multi-area OSPF enterprise network topology for KtmTechnologies, designed in Cisco Packet Tracer.

2.1 OSPF Area Plan

Area	**Member Networks**
Area 0 (Backbone)	Border and core routers, Data Center Server LAN, and Guest LAN
Area 1	HQ Corporate VLANs (Finance, HR and Legal), Sales, and Marketing
Area 2	R&D, Production Floor, Warehouse & Logistics, and Branch Office

2.2 Router and LAN Summary

Device	**Area**	**Router Connections**	**LAN(s) Served**
ISP-R1	ISP	BR (P01)	ISP DNS LAN
BR	0	ISP, CR1, CR2 (P01–P03)	None; transit router
CR1	0	BR, CR2, R11, R5, R4	None; transit/core router
CR2	0	BR, CR1, R11, R5, R4, R6	None; transit/core router
R11	0	CR1, CR2 (P05, P06)	Data Center and Guest LAN
R4	1	CR1, CR2, R8 (P09–P11)	Finance, HR and Legal VLANs
R8	1	R4 (P11)	Sales and Marketing
R5	2	CR1, CR2 (P07, P08)	R&D Lab
R6	2	CR2, R7 (P12, P13)	Production Floor
R7	2	R6, R9 (P13, P14)	Warehouse & Logistics
R9	2	R7 (P14)	Branch Office

3 Addressing Plan

3.1 IP Address Pool

- **Private enterprise block:** `10.0.0.0/16`
- **Allocated LAN/VLAN range:** `10.0.0.0` – `10.0.6.207`
- **Internal point-to-point pool:** `10.0.12.0/24`
- **Reserved growth space:** `10.0.7.0` – `10.0.11.255`, and `10.0.13.0` onward
- **ISP documentation blocks:** `198.51.100.0/30` (BR–ISP) and `198.51.100.4/30` (ISP DNS LAN)

3.2 LAN and VLAN Subnets

The eleven LAN/VLAN subnets use six prefix sizes (/23, /24, /25, /26, /27 and /28). They are allocated from largest to smallest to reduce fragmentation.

Sn	**Network**	**Req.**	**Alloc.**	**Subnet Mask**	**Network ID**	**Usable Range**
1	Production Floor	500	510	255.255.254.0	`10.0.0.0/23`	.0.1–.1.254
2	R&D Lab	250	254	255.255.255.0	`10.0.2.0/24`	.2.1–.2.254
3	Warehouse & Logistics	200	254	255.255.255.0	`10.0.3.0/24`	.3.1–.3.254
4	Branch Office	150	254	255.255.255.0	`10.0.4.0/24`	.4.1–.4.254
5	Sales	100	126	255.255.255.128	`10.0.5.0/25`	.5.1–.5.126
6	Marketing	80	126	255.255.255.128	`10.0.5.128/25`	.5.129–.5.254
7	Data Center / Server LAN	60	62	255.255.255.192	`10.0.6.0/26`	.6.1–.6.62
8	Guest LAN	40	62	255.255.255.192	`10.0.6.64/26`	.6.65–.6.126
9	Finance VLAN	30	30	255.255.255.224	`10.0.6.128/27`	.6.129–.6.158
10	HR VLAN	25	30	255.255.255.224	`10.0.6.160/27`	.6.161–.6.190
11	Legal VLAN	10	14	255.255.255.240	`10.0.6.192/28`	.6.193–.6.206

3.3 Point-to-Point Links

Each link uses a /30 subnet with two usable addresses. The internal links are taken sequentially from `10.0.12.0/24`. The P01 link between the Border Router (BR) and the ISP router uses `198.51.100.0/30`, a documentation block reserved by RFC 5737. It represents the public-facing boundary between the ISP and the private `10.0.0.0/16` enterprise network. In a real deployment, the ISP would provide a public IP address and BR would use NAT before internal traffic leaves the network.

Link	Network ID	Usable Host IPs	Purpose
P01	198.51.100.0/30	198.51.100.1, 198.51.100.2	ISP-R1–BR
P02	10.0.12.0/30	10.0.12.1, 10.0.12.2	BR–CR1
P03	10.0.12.4/30	10.0.12.5, 10.0.12.6	BR–CR2
P04	10.0.12.8/30	10.0.12.9, 10.0.12.10	CR1–CR2 core loop
P05	10.0.12.12/30	10.0.12.13, 10.0.12.14	CR1–R11
P06	10.0.12.16/30	10.0.12.17, 10.0.12.18	CR2–R11 redundant
P07	10.0.12.20/30	10.0.12.21, 10.0.12.22	CR1–R5
P08	10.0.12.24/30	10.0.12.25, 10.0.12.26	CR2–R5 redundant
P09	10.0.12.28/30	10.0.12.29, 10.0.12.30	CR1–R4
P10	10.0.12.32/30	10.0.12.33, 10.0.12.34	CR2–R4 redundant
P11	10.0.12.36/30	10.0.12.37, 10.0.12.38	R4–R8
P12	10.0.12.40/30	10.0.12.41, 10.0.12.42	CR2–R6
P13	10.0.12.44/30	10.0.12.45, 10.0.12.46	R6–R7 plant backbone
P14	10.0.12.48/30	10.0.12.49, 10.0.12.50	R7–R9 branch link

4 Server and Service Placement

Server	IP	Location	Purpose
Web Server 1	10.0.6.10	Data Center, Area 0	Hosts the company website
DNS Server 1	10.0.6.11	Data Center, Area 0	Resolves the company's domain names
DNS Server 2	10.0.4.10	Branch, Area 2	Backup DNS in case DNS 1 goes down
Web Server 2	10.0.2.10	R&D, Area 2	Backup copy of the website in case Web 1 goes down
ISP DNS Server	198.51.100.6	ISP LAN	Resolves domain names outside the company network
DHCP Server (Data Center)	10.0.6.2	Data Center, Area 0	Assigns addresses to the Data Center LAN
DHCP Server (Guest)	10.0.6.66	Guest LAN, Area 0	Assigns addresses to the Guest LAN
DHCP Server (HQ)	10.0.6.130	HQ, Area 1	Assigns addresses to the HQ VLANs
DHCP Server (Sales)	10.0.5.2	Sales, Area 1	Assigns addresses to the Sales LAN
DHCP Server (Marketing)	10.0.5.130	Marketing, Area 1	Assigns addresses to the Marketing LAN
DHCP Server (R&D)	10.0.2.2	R&D, Area 2	Assigns addresses to the R&D LAN
DHCP Server (Production)	10.0.0.2	Production, Area 2	Assigns addresses to the Production LAN
DHCP Server (Warehouse)	10.0.3.2	Warehouse, Area 2	Assigns addresses to the Warehouse LAN
DHCP Server (Branch)	10.0.4.2	Branch, Area 2	Assigns addresses to the Branch LAN

5 Conclusion

The proposed KtmTechnologies network meets the stated minimum requirements of Mini-Project: Computer Networks, while remaining practical to implement in Cisco Packet Tracer. Its VLSM plan uses addresses efficiently, multi-area OSPF organizes routing, and the redundant core improves availability. The design is ready for detailed interface configuration, end-device placement, DHCP/DNS setup and connectivity testing.